

Design and Implementation of a Highly Available Three-Tier Web Architecture on Amazon Web Services (AWS)

Project Report

INTRODUCTION

1.1 Project Overview

Cloud computing has transformed the way organizations deploy and manage applications by providing scalable, reliable, and cost-effective infrastructure. Amazon Web Services (AWS) is one of the leading cloud service providers that offers a comprehensive suite of services for building secure and highly available applications.

This project demonstrates the implementation of a Three-Tier Web Architecture on AWS using a custom Virtual Private Cloud (VPC), Auto Scaling Group (ASG), Application Load Balancer (ALB), Bastion Host, and EC2 instances. The infrastructure is designed to provide high availability, scalability, fault tolerance, and secure access by distributing resources across multiple Availability Zones.

The project follows AWS best practices by separating the network into public and private subnets. Administrative access is provided through a Bastion Host, while the web servers remain isolated within private subnets. Traffic from end users is managed by an Application Load Balancer, ensuring efficient load distribution among backend servers.

1.2 Objectives

- Design a secure AWS Virtual Private Cloud.
- Deploy highly available web servers across multiple Availability Zones.
- Implement Auto Scaling for automatic resource management.
- Configure Application Load Balancer for traffic distribution.
- Improve infrastructure security using private subnets.
- Study AWS networking and security services.

1.3 Scope of the Project

The project focuses on deploying a production-style cloud infrastructure suitable for hosting scalable web applications. It includes networking, compute, load balancing, security, and automated scaling services available in AWS.

1.4 AWS Services Used

- Amazon VPC
- Amazon EC2
- Security Groups
- Launch Template
- Auto Scaling Group

- Application Load Balancer
- Target Group
- NAT Gateway
- Internet Gateway
- Bastion Host
- CloudWatch (Concept)
- IAM (Basic)

1.5 Three-Tier Architecture Overview

A Three-Tier Architecture separates the application into three logical layers:

- **Presentation Tier:** Receives client requests through the Application Load Balancer.
- **Application Tier:** EC2 instances process the client requests.
- **Data Tier:** Future implementation can include Amazon RDS for database services.

This separation improves security, scalability, maintainability, and performance.

2. SYSTEM DESIGN AND ARCHITECTURE

2.1 Architecture Overview

The implemented architecture consists of one custom VPC with two public subnets and two private subnets distributed across two Availability Zones.

Public subnets contain the Bastion Host, NAT Gateway, and Application Load Balancer, while private subnets host the web server instances created through the Auto Scaling Group.

This design ensures that backend servers remain inaccessible directly from the Internet.

2.2 Virtual Private Cloud (VPC)

The Sakura VPC provides a logically isolated network where all cloud resources are deployed.

S.no.	Configuration	Value
1	VPC Name	Sakura
2	IPv4 CIDR	10.0.0.0/16
3	Availability Zones	2
4	Public Subnets	2
5	Private Subnets	2

Advantages

- Network isolation
- Custom IP addressing

- Secure communication
- Flexible routing
- High Availability

2.3 NAT Gateway Analysis

The NAT Gateway enables instances located in private subnets to access the Internet for downloading updates and software packages while preventing inbound Internet access.

Zonal NAT Gateway

Advantages

- High Availability
- No cross-AZ dependency
- Better fault tolerance

Disadvantages

- Higher monthly cost

Regional NAT Gateway

Advantages

- Lower deployment cost

Disadvantages

- Cross-AZ traffic charges
- Possible single point of failure

For production deployments, one NAT Gateway should be deployed in each Availability Zone to eliminate single points of failure.

2.4 VPC Endpoint Analysis

Two configurations were studied.

Without VPC Endpoint

Private EC2 → NAT Gateway → Internet → Amazon S3

Using S3 Gateway Endpoint

Private EC2 → S3 Gateway Endpoint → Amazon S3

Benefits

- Lower data transfer cost
- Better security
- Faster communication
- Reduced NAT Gateway usage

2.5 Security Group

The Security Group acts as a virtual firewall controlling inbound and outbound traffic.

S.no.	Port	Purpose
1	22	SSH Access
2	8000	Web Application

In production, SSH access should be restricted to trusted IP addresses instead of allowing traffic from anywhere.

3. IMPLEMENTATION AND CONFIGURATION

A Launch Template defines a standard configuration for EC2 instances.

Configuration includes:

- Ubuntu AMI
- t3.micro
- Sakura Key Pair
- Security Group
- Storage Configuration

Using Launch Templates ensures consistency across all instances launched by the Auto Scaling Group.

3.2 Bastion Host

A Bastion Host is deployed inside the public subnet to securely access EC2 instances located within private subnets.

Advantages

- Centralized administration
- Enhanced security
- Private instances remain inaccessible from the Internet

3.3 Auto Scaling Group

Configuration

Desired Capacity: 2

Minimum Capacity: 1

Maximum Capacity: 4

The Auto Scaling Group automatically maintains the desired number of EC2 instances and replaces unhealthy instances.

3.4 Application Load Balancer

The Application Load Balancer distributes incoming HTTP traffic among healthy backend servers.

Benefits

- High Availability
- Fault Tolerance
- Load Distribution
- Automatic Health Checks

3.5 Target Group

The Target Group registers backend EC2 instances and continuously monitors their health before forwarding traffic.

3.6 Linux Configuration

Commands used:

- `chmod 400 sakura.pem` - Restricts private key permissions.
- `ssh -i sakura.pem ubuntu@Private-IP` - Connects to the private EC2 instance.
- `vim index.html` - Edits the web page.
- `cat index.html` - Displays file contents.

4. ANALYSIS AND DISCUSSION

4.1 High Availability

Deploying resources across two Availability Zones ensures uninterrupted service during hardware or Availability Zone failures.

4.2 Security Analysis

Security is enhanced by:

- Private Subnets
- Bastion Host
- Security Groups
- NAT Gateway
- Network Isolation

Future improvements include:

- AWS WAF
- AWS Shield
- Session Manager

- IAM Roles

4.3 Cost Analysis

S.no.	Service	Estimated Cost Impact
1	EC2 t3.micro	Low
2	ALB	Medium
3	NAT Gateway	High
4	VPC	Free
5	Security Groups	Free
6	Auto Scaling	No Extra charge

Using S3 Gateway Endpoints can significantly reduce NAT Gateway costs.

4.4 Performance Analysis

Auto Scaling improves performance by automatically launching additional instances during periods of high CPU utilization.

Recommended CPU Threshold

Scale Out: 70%

Scale In: 30%

4.5 Best Practices

- Use Multi-AZ deployment.
- Deploy one NAT Gateway per AZ.
- Restrict SSH access.
- Enable HTTPS using ACM.
- Monitor infrastructure using CloudWatch.
- Use IAM Roles instead of access keys.
- Enable automatic backups.

5.Conclusion

The Sakura project successfully demonstrates the implementation of a secure and highly available Three-Tier Web Architecture using AWS cloud services. The project integrates networking, security, compute, and load balancing services into a scalable infrastructure capable of handling varying workloads.

By implementing a custom VPC, Launch Templates, Auto Scaling Groups, Application Load Balancers, and secure Bastion Host access, the architecture follows modern cloud computing best practices while ensuring scalability, fault tolerance, and improved security.

The project provides practical knowledge of AWS networking and infrastructure deployment and serves as a strong foundation for deploying enterprise-grade cloud applications.

